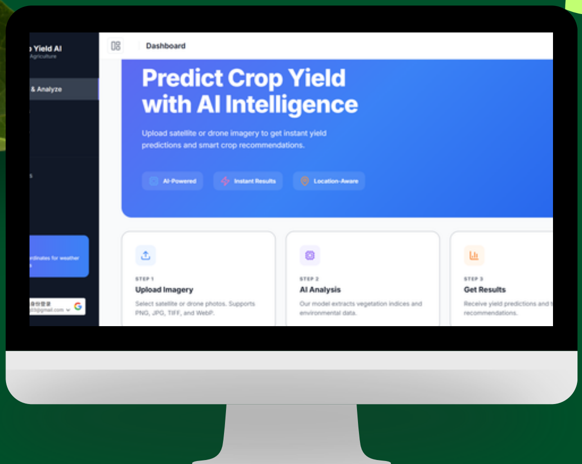
About the Project

A WEB-BASED AI PLATFORM FOR LAND SUITABILITY ASSESSMENT, CROP YIELD PREDICTION, CROP RECOMMENDATION, AND ENVIRONMENTAL INSIGHTS.

System Preview



Problem Statement

- LAND ASSESSMENT IS COSTLY AND SLOW.
- FARMERS MAY LACK RELIABLE LAND INFORMATION.
- AI CAN ASSESS LAND FASTER USING IMAGERY AND ENVIRONMENTAL DATA.

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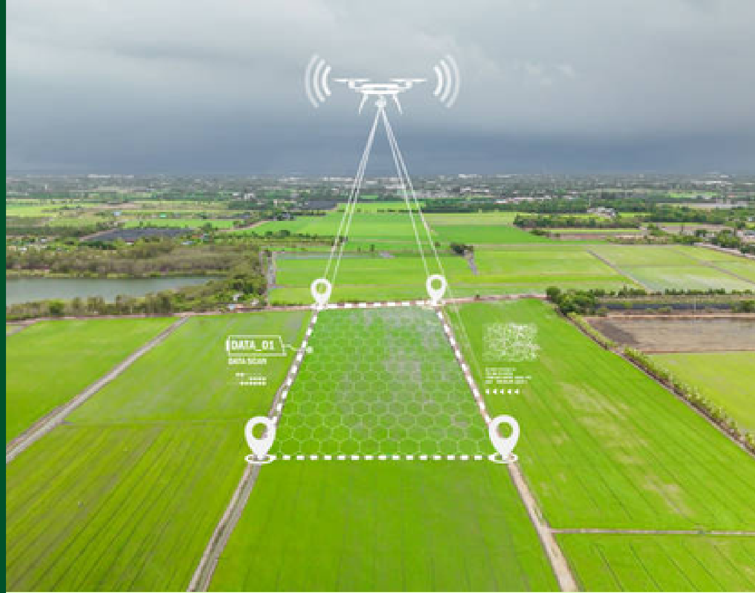

Contact

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https://crop-yield-ygmx.onrender.com

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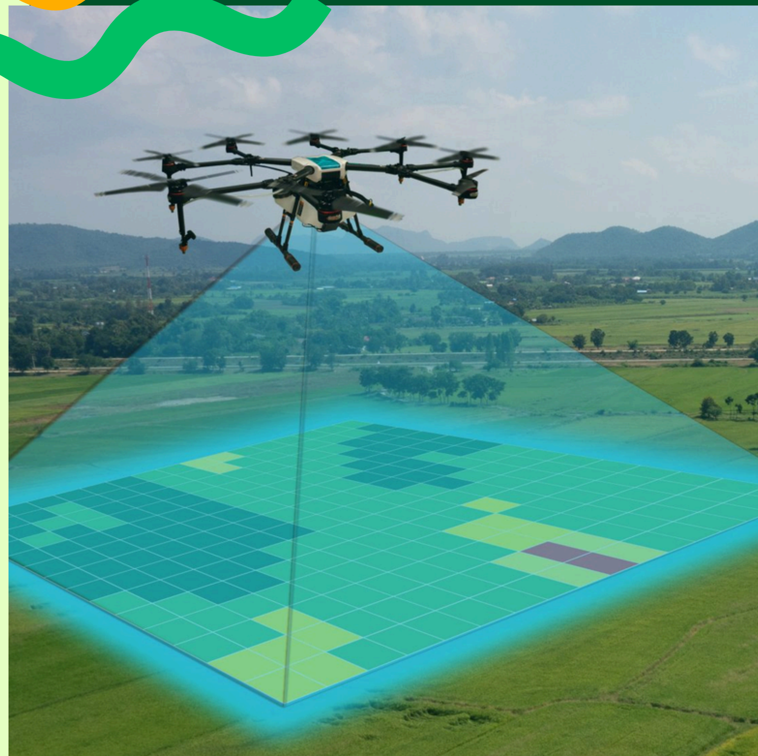
AI-Powered Crop Yield Prediction

Smart agriculture for better land decisions.



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Objectives

1. DEVELOP AN AI-POWERED SYSTEM TO ASSESS AGRICULTURAL LAND SUITABILITY USING REMOTE SENSING AND ENVIRONMENTAL DATA TO AID IN DECISION-MAKING.
2. RECOMMEND THE TOP THREE MOST SUITABLE OR HIGHEST YIELDING CROPS FOR A GIVEN LAND PLOT BASED ON THE PREDICTIONS AND ENVIRONMENTAL FACTORS.
3. EVALUATE THE PERFORMANCE OF THE DEVELOPED SYSTEM USING REGRESSION AND CLASSIFICATION METRICS FOR THE QUALITY OF MODEL.

System Features

- IMAGE UPLOAD AND ANALYSIS
- CROP YIELD PREDICTION
- LAND SUITABILITY CLASSIFICATION
- TOP 3 RECOMMENDATIONS
- ENVIROMENT SNAPSHOT
- LOCATION-BASED UPDATE

Methodology

Our system follows a structured workflow to transform raw data into accurate predictions and actionable recommendations.

1. **DATA COLLECTION**
   Collect satellite, drone, and environmental data with location information.
2. **IMAGE PREPROCESSING**
   Clean, resize, and normalize images for consistent analysis.
3. **MODEL DEVELOPMENT**
   Train and evaluate machine learning models for yield and suitability prediction.
4. **SYSTEM INTEGRATION**
   Integrate models and data pipelines into a web-based platform.
5. **USABILITY TESTING**
   Evaluate usability and refine the system based on user feedback.

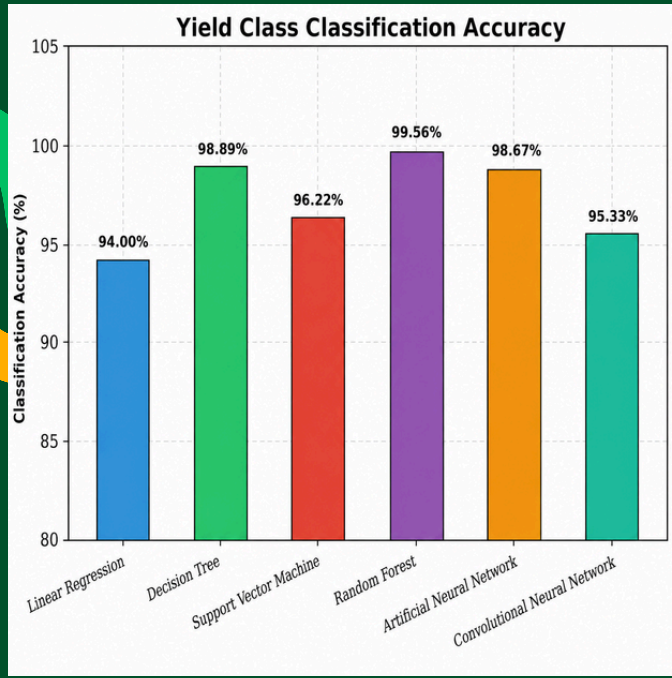
Datasets

Agricultural images were collected from Kaggle, satellite imagery, drone imagery, and environmental sources.

Tech Stack

- PYTHON
- SUPABASE
- WEB PLATFORM
- AI / MACHINE LEARNING

Model Result



RANDOM FOREST WAS SELECTED BECAUSE IT GAVE BALANCED RESULTS FOR CROP YIELD PREDICTION AND LAND SUITABILITY CLASSIFICATION.

Usability Feedback

USABILITY TESTING SHOWED THAT THE SYSTEM WAS EASY TO UNDERSTAND, SIMPLE TO NAVIGATE, AND USEFUL FOR VIEWING PREDICTION RESULTS.